


```

31         printf("echidna, the spiny anteater\n");
32         break;
33     case 'f' :
34         printf("fisher, brownish marten\n");
35         break;
36     default :
37         printf("That's a stumper!\n");
38     } /* end of switch */
39     else
40         printf("I recognize only lowercase letters.\n");
41     while (getchar() != '\n')
42         continue; /* skip rest of input line */
43     printf("Please type another letter or a #.\n");
44 } /* while loop end */
45 printf("Bye!\n");
46
47 return 0;
48 }

```

```

1 gcc animals.c -o animals && echo "a`nb`nc`n#`n" | ./animals
2 Give me a letter of the alphabet, and I will give an animal name
3 beginning with that letter.
4 Please type in a letter; type # to end my act.
5 argali, a wild sheep of Asia
6 Please type another letter or a #.
7 babirusa, a wild pig of Malay
8 Please type another letter or a #.
9 coati, racoonlike mammal
10 Please type another letter or a #.
11 Bye!

```

break.c

- 作用: 演示使用 `break` 语句在输入条件不满足时退出循环。
- 功能: 程序从标准输入依次读取矩形的长度和宽度，计算并输出面积。当无法读取宽度时（如输入非数字）结束循环。
- 演示效果:

```

1 /* break.c -- uses break to exit a loop */
2 #include <stdio.h>
3 int main(void)

```

```

4  {
5      float length, width;
6
7      printf("Enter the length of the rectangle:\n");
8      while (scanf("%f", &length) == 1)
9      {
10         printf("Length = %0.2f:\n", length);
11         printf("Enter its width:\n");
12         if (scanf("%f", &width) != 1)
13             break;
14         printf("Width = %0.2f:\n", width);
15         printf("Area = %0.2f:\n", length * width);
16         printf("Enter the length of the rectangle:\n");
17     }
18     printf("Done.\n");
19
20     return 0;
21 }

```

```

1  gcc break.c -o break && echo "5`n3`n7.5`n2.2`na`n" | ./break
2  Enter the length of the rectangle:
3  Length = 5.00:
4  Enter its width:
5  Width = 3.00:
6  Area = 15.00:
7  Enter the length of the rectangle:
8  Length = 7.50:
9  Enter its width:
10 Width = 2.20:
11 Area = 16.50:
12 Enter the length of the rectangle:
13 Done.

```

chcount.c

- 作用: 演示逻辑与操作符 `&&` 的使用, 过滤特定字符计数。
- 功能: 程序从输入读取字符直到遇到句号 `.` 为止, 统计除双引号 `"` 和单引号 `'` 外的字符数量。
- 演示效果:

```

1  // chcount.c -- use the logical AND operator
2  #include <stdio.h>

```

```

3  #define PERIOD '.'
4  int main(void)
5  {
6      char ch;
7      int charcount = 0;
8
9      while ((ch = getchar()) != PERIOD)
10     {
11         if (ch != '"' && ch != '\\')
12             charcount++;
13     }
14     printf("There are %d non-quote characters.\n", charcount);
15
16     return 0;
17 }

```

```

1  gcc chcount.c -o chcount && echo "He said \"Hello\".\`n" | ./chcount
2  There are 16 non-quote characters.

```

colddays.c

- 作用: 统计低于冰点（摄氏0度）的天数百分比。
- 功能: 程序从输入读取每天的最低温度，统计总天数以及低于冰点的天数，最后输出百分比。
- 演示效果:

```

1  // colddays.c -- finds percentage of days below freezing
2  #include <stdio.h>
3  int main(void)
4  {
5      const int FREEZING = 0;
6      float temperature;
7      int cold_days = 0;
8      int all_days = 0;
9
10     printf("Enter the list of daily low temperatures.\n");
11     printf("Use Celsius, and enter q to quit.\n");
12     while (scanf("%f", &temperature) == 1)
13     {
14         all_days++;
15         if (temperature < FREEZING)
16             cold_days++;

```

```

17     }
18     if (all_days != 0)
19         printf("%d days total: %.1f%% were below freezing.\n",
20             all_days, 100.0 * (float) cold_days / all_days);
21     if (all_days == 0)
22         printf("No data entered!\n");
23
24     return 0;
25 }

```

```

1 gcc colddays.c -o colddays && echo "-5`n3`n0`n-2`nq`n" | ./colddays
2 Enter the list of daily low temperatures.
3 Use Celsius, and enter q to quit.
4 4 days total: 50.0% were below freezing.

```

cypher1.c

- 作用: 将输入的每个非空格字符 ASCII 编码加 1, 空格保持不变。
- 功能: 程序读取一行输入, 将空格原样输出, 其它字符的 ASCII 值加 1 后输出, 最后换行结束。
- 演示效果:

```

1 // cypher1.c -- alters input, preserving spaces
2 #include <stdio.h>
3 #define SPACE ' ' // that's quote-space-quote
4 int main(void)
5 {
6     char ch;
7
8     ch = getchar(); // read a character
9     while (ch != '\n') // while not end of line
10     {
11         if (ch == SPACE) // leave the space
12             putchar(ch); // character unchanged
13         else
14             putchar(ch + 1); // change other characters
15         ch = getchar(); // get next character
16     }
17     putchar(ch); // print the newline
18
19     return 0;

```

```

1 gcc cypher1.c -o cypher1 && echo "Hello World`n" | ./cypher1
2 Ifmmp Xpsme

```

cypher2.c

- 作用: 将输入的字母字符替换为 ASCII 编码加 1 的新字符, 其他非字母字符保持原样。
- 功能: 程序逐个读取字符, 若是字母则输出其下一个字母 (字符码加1), 否则原样输出。
- 演示效果:

```

1 // cypher2.c -- alters input, preserving non-letters
2 #include <stdio.h>
3 #include <ctype.h>           // for isalpha()
4 int main(void)
5 {
6     char ch;
7
8     while ((ch = getchar()) != '\n')
9     {
10         if (isalpha(ch))      // if a letter,
11             putchar(ch + 1);  // display next letter
12         else                  // otherwise,
13             putchar(ch);      // display as is
14     }
15     putchar(ch);              // display the newline
16
17     return 0;
18 }

```

```

1 gcc cypher2.c -o cypher2 && echo "Hello 123!`n" | ./cypher2
2 Ifmmp 123!

```

divisors.c

- 作用: 检测一个整数并输出它的约数, 如果是质数则标记。
- 功能: 程序读取整数并检查它的所有可能的约数, 若能整除则输出配对约数, 否则标记为质数; 支持连续输入直到退出。

- 演示效果:

```
1 // divisors.c -- nested ifs display divisors of a number
2 #include <stdio.h>
3 #include <stdbool.h>
4 int main(void)
5 {
6     unsigned long num;          // number to be checked
7     unsigned long div;          // potential divisors
8     bool isPrime;              // prime flag
9
10    printf("Please enter an integer for analysis; ");
11    printf("Enter q to quit.\n");
12    while (scanf("%lu", &num) == 1)
13    {
14        for (div = 2, isPrime = true; (div * div) ≤ num; div++)
15        {
16            if (num % div == 0)
17            {
18                if ((div * div) ≠ num)
19                    printf("%lu is divisible by %lu and %lu.\n",
20                        num, div, num / div);
21                else
22                    printf("%lu is divisible by %lu.\n",
23                        num, div);
24                isPrime = false; // number is not prime
25            }
26        }
27        if (isPrime)
28            printf("%lu is prime.\n", num);
29        printf("Please enter another integer for analysis; ");
30        printf("Enter q to quit.\n");
31    }
32    printf("Bye.\n");
33
34    return 0;
35 }
```

```

1 gcc divisors.c -o divisors && echo "28`n13`nq`n" | ./divisors
2 Please enter an integer for analysis; Enter q to quit.
3 28 is divisible by 2 and 14.
4 28 is divisible by 4 and 7.
5 Please enter another integer for analysis; Enter q to quit.
6 13 is prime.
7 Please enter another integer for analysis; Enter q to quit.
8 Bye.

```

electric.c

- 作用: 根据分段电价计算用电费用。
- 功能: 程序获取输入的用电量（千瓦时），根据不同用电区间使用对应电价计算总费用。
- 演示效果:

```

1 // electric.c -- calculates electric bill
2 #include <stdio.h>
3 #define RATE1 0.13230 // rate for first 360 kwh
4 #define RATE2 0.15040 // rate for next 108 kwh
5 #define RATE3 0.30025 // rate for next 252 kwh
6 #define RATE4 0.34025 // rate for over 720 kwh
7 #define BREAK1 360.0 // first breakpoint for rates
8 #define BREAK2 468.0 // second breakpoint for rates
9 #define BREAK3 720.0 // third breakpoint for rates
10 #define BASE1 (RATE1 * BREAK1)
11 // cost for 360 kwh
12 #define BASE2 (BASE1 + (RATE2 * (BREAK2 - BREAK1)))
13 // cost for 468 kwh
14 #define BASE3 (BASE1 + BASE2 + (RATE3 * (BREAK3 - BREAK2)))
15 //cost for 720 kwh
16 int main(void)
17 {
18     double kwh; // kilowatt-hours used
19     double bill; // charges
20
21     printf("Please enter the kwh used.\n");
22     scanf("%lf", &kwh); // %lf for type double
23     if (kwh ≤ BREAK1)
24         bill = RATE1 * kwh;
25     else if (kwh ≤ BREAK2) // kwh between 360 and 468
26         bill = BASE1 + (RATE2 * (kwh - BREAK1));

```



```

27     else if (kwh ≤ BREAK3)    // kwh between 468 and 720
28         bill = BASE2 + (RATE3 * (kwh - BREAK2));
29     else                        // kwh above 680
30         bill = BASE3 + (RATE4 * (kwh - BREAK3));
31     printf("The charge for %.1f kwh is $%1.2f.\n", kwh, bill);
32
33     return 0;
34 }

```

```

1 gcc electric.c -o electric && echo "800`n" | ./electric
2 Please enter the kwh used.
3 The charge for 800.0 kwh is $214.38.

```

paint.c

- 作用: 根据待涂面积计算所需油漆桶数, 使用条件运算符简化判断。
- 功能: 程序每次读取待涂面积, 计算所需油漆桶数 (每桶覆盖 350 平方英尺), 考虑余数决定是否需
要额外一桶, 循环直到退出。
- 演示效果:

```

1  /* paint.c -- uses conditional operator */
2  #include <stdio.h>
3  #define COVERAGE 350           // square feet per paint can
4  int main(void)
5  {
6      int sq_feet;
7      int cans;
8
9      printf("Enter number of square feet to be painted:\n");
10     while (scanf("%d", &sq_feet) == 1)
11     {
12         cans = sq_feet / COVERAGE;
13         cans += ((sq_feet % COVERAGE == 0)) ? 0 : 1;
14         printf("You need %d %s of paint.\n", cans,
15              cans == 1 ? "can" : "cans");
16         printf("Enter next value (q to quit):\n");
17     }
18
19     return 0;
20 }

```

```
1 gcc paint.c -o paint && echo "400`n300`nq`n" | ./paint
2 Enter number of square feet to be painted:
3 You need 2 cans of paint.
4 Enter next value (q to quit):
```

skippart.c

- 作用: 演示使用 `continue` 跳过循环中的部分逻辑。
- 功能: 程序读取分数并验证其在有效范围 [0,100], 不合法则跳过当前循环, 不计入统计; 合法分数记录最小值、最大值并计算平均值。
- 演示效果:

```
1  /* skippart.c  -- uses continue to skip part of loop */
2  #include <stdio.h>
3  int main(void)
4  {
5      const float MIN = 0.0f;
6      const float MAX = 100.0f;
7
8      float score;
9      float total = 0.0f;
10     int n = 0;
11     float min = MAX;
12     float max = MIN;
13
14     printf("Enter the first score (q to quit): ");
15     while (scanf("%f", &score) == 1)
16     {
17         if (score < MIN || score > MAX)
18         {
19             printf("%.1f is an invalid value. Try again: ",
20                 score);
21             continue; // jumps to while loop test condition
22         }
23         printf("Accepting %.1f:\n", score);
24         min = (score < min)? score: min;
25         max = (score > max)? score: max;
26         total += score;
27         n++;
28         printf("Enter next score (q to quit): ");
29     }
```

```

30     if (n > 0)
31     {
32         printf("Average of %d scores is %0.1f.\n", n, total / n);
33         printf("Low = %0.1f, high = %0.1f\n", min, max);
34     }
35     else
36         printf("No valid scores were entered.\n");
37     return 0;
38 }

```

```

1  gcc skippart.c -o skippart && echo "50`n120`n80`n-5`n70`nq`n" | ./skippart
2  Enter the first score (q to quit): Accepting 50.0:
3  Enter next score (q to quit): 120.0 is an invalid value. Try again: Accepting
   80.0:
4  Enter next score (q to quit): -5.0 is an invalid value. Try again: Accepting
   70.0:
5  Enter next score (q to quit): Average of 3 scores is 66.7.
6  Low = 50.0, high = 80.0

```

vowels.c

- 作用: 统计输入文本中各元音字母 (A、E、I、O、U) 的出现次数。
- 功能: 程序读取输入直到遇到 `#` 结束, 针对大小写元音字母使用多标签 `switch` 统计出现次数并输出结果。
- 演示效果:

```

1  // vowels.c -- uses multiple labels
2  #include <stdio.h>
3  int main(void)
4  {
5      char ch;
6      int a_ct, e_ct, i_ct, o_ct, u_ct;
7
8      a_ct = e_ct = i_ct = o_ct = u_ct = 0;
9
10     printf("Enter some text; enter # to quit.\n");
11     while ((ch = getchar()) != '#')
12     {
13         switch (ch)
14         {
15             case 'a' :

```

```

16         case 'A' : a_ct++;
17             break;
18         case 'e' :
19         case 'E' : e_ct++;
20             break;
21         case 'i' :
22         case 'I' : i_ct++;
23             break;
24         case 'o' :
25         case 'O' : o_ct++;
26             break;
27         case 'u' :
28         case 'U' : u_ct++;
29             break;
30         default : break;
31     }
32     // end of switch
33 }
34 // while loop end
35 printf("number of vowels:  A    E    I    O    U\n");
36 printf("                    %4d %4d %4d %4d %4d\n",
37         a_ct, e_ct, i_ct, o_ct, u_ct);
38
39 return 0;
40 }

```

```

1 gcc vowels.c -o vowels && echo "Hello World AEIOU aeio#\n" | ./vowels
2 Enter some text; enter # to quit.
3 number of vowels:  A    E    I    O    U
4                   2    3    2    4    1

```

wordcnt.c

- 作用: 统计输入文本的字符数、单词数、行数以及未完整结束的行数。
- 功能: 程序逐字符读取输入，识别单词边界与换行符，并在输入以 `|` 符号终止时输出统计结果。
- 演示效果:

```

1 // wordcnt.c -- counts characters, words, lines
2 #include <stdio.h>
3 #include <ctype.h>      // for isspace()
4 #include <stdbool.h>    // for bool, true, false
5 #define STOP '|'
6 int main(void)

```

```

7  {
8      char c;                // read in character
9      char prev;             // previous character read
10     long n_chars = 0L;      // number of characters
11     int n_lines = 0;         // number of lines
12     int n_words = 0;         // number of words
13     int p_lines = 0;         // number of partial lines
14     bool inword = false;     // = true if c is in a word
15
16     printf("Enter text to be analyzed (| to terminate):\n");
17     prev = '\n';             // used to identify complete lines
18     while ((c = getchar()) != STOP)
19     {
20         n_chars++;           // count characters
21         if (c == '\n')
22             n_lines++;       // count lines
23         if (!isspace(c) && !inword)
24         {
25             inword = true;   // starting a new word
26             n_words++;        // count word
27         }
28         if (isspace(c) && inword)
29             inword = false;  // reached end of word
30         prev = c;            // save character value
31     }
32
33     if (prev != '\n')
34         p_lines = 1;
35     printf("characters = %ld, words = %d, lines = %d, ",
36           n_chars, n_words, n_lines);
37     printf("partial lines = %d\n", p_lines);
38
39     return 0;
40 }

```

```

1  gcc wordcnt.c -o wordcnt && echo "Hello world\nThis is C language|\n" |
   ./wordcnt
2  Enter text to be analyzed (| to terminate):
3  characters = 31, words = 5, lines = 0, partial lines = 1

```

You need 1 can of paint.

Enter next value (q to quit):

1 |

1 |

1 |

1 |